

line might be built out of chips that drive today's consumer electronics equipments like iPods and mobile phones.

Researchers at the Lawrence Berkeley National Laboratory, a US Department of Energy lab have, in a research paper proposed the building of a poor man's supercomputer using the embedded chips. Meant to be used for running cloud simulations, these systems are used in weather forecasting and rain predictions. Currently, the fastest supercomputers can do this. However, the accuracy level and leaves much to be desired which leads the Met department's weather forecasts to be taken with a healthy pinch of salt. With this new "iPod supercomputer", however, scientists hope that predicting the vagaries of the weather would be a lot easier.

In the midst of all this, the interesting fact is the saving in money as a result of all this design. A conventional supercomputer built with the existing design would cost around \$1 Billion and consume around 200 MW of electricity, which is enough to power a city of around 100,000 people. This supercomputer would also need to be more than 1,000 times more powerful than the current models. The new design would use 20 million chips, of the kind that is found in iPods. In addition, it would also cost a relatively less \$75million. Electric power consumption would also considerably be less at 4 MW, and performance figures would be 200 petaflops (IBM's Blue Gene, one of the current fastest runs at 2 petaflops).

The only hurdle appears to be the interconnection between individual processors—the paths through which data flows. If there are any bottlenecks, then the whole scheme would come crashing down. We guess researchers would surely find one solution or another.

PREACHING VIA SMS

Teh V471c4n, And Teh Pope?

The Vatican seems to be finally waking up to the axiom that when you can't take a horse to the river, bring the river to the horse. Hit by falling rates of attendance at churches and the general disdain of the younger generation towards God and institutional religion, the Vatican is hoping that technology can help them reach out to the young.

In the run up to World Youth Day in July, Pope Benedict XVI will text prayers and blessings to thousands of Catholic youth. To be held at Sydney, and spread over six days, the World Youth Day is an event promoted by the Roman Catholic church, which basically tries to "catch 'em young"! Thousands of young men and women from across the world gather together every two or three years to meet each other and talk about spirituality and religion. This audience, accustomed to mobile phones and the Internet are believed to be better reached, if the traditional sermons were replaced with something more techie. Besides using text messages, the church also plans to promote a social networking site akin to Orkut or Facebook to knit the community, especially the youth together. Digital prayer walls would also be erected across the sprawling racecourse in Sydney, where the main events are planned, so that participants can see the prayers instead of straining to hear from the nearest loudspeaker.

The telecom company in charge of these arrangements estimates that it will connect 8,000 volunteers, 2,000 members of the clergy, 3,000 media personnel and about 225,000 pilgrims across 700 locations in Sydney. The faithful who can't make it to Sydney will



Femtocells

What are Femtocells?

Femtocells, officially known as Access Point Base Stations are small devices that boost cellular networks around areas in the interior of buildings that might have low signal strength. A femtocell is no different from a wireless router and are about the size of a paperback.

How do they work?

Femto cells route the signals from up to 5 mobile phones through a broadband connection and pass them to the service provider thereby bypassing the conventional phone to tower path. GSM, CDMA and UMTS connections are currently supported by femtocells.

Who will benefit?

Femtocells benefit both customers and operators. Customers benefit by improved coverage, lower costs per call, longer battery life and usage of multiple phones and lines on the same connection. Operators get more customers by minimal infrastructure investment and reduced congestion in the wireless network. Therefore it's a win-win scenario for everyone involved.

Why the emphasis on Femtocells?

With the phasing out of 2G networks, phones using 3G networks and technologies like HSDPA becoming mainstream, extending a network by conventional means—installing more towers—would be expensive. Also since people are used to terms like Internet and broadband, customer education would not be necessary and adaptation to the new technology would be hassle-free. The technological advances in embedded design have also lowered the price of base stations to affordable levels.

When will this technology debut?

Femtocells are already used on a limited basis across some cities in the West.

Which companies are working on this?

Major players like Motorola, Netgear and Google have their fingers in the femtocell pie.